Dummy Coding

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Introduction

Categorical predictors in regression must be encoded as numeric indicator variables before they enter the linear predictor. R's default is treatment-contrast coding (also called dummy coding), where one level is the implicit reference and the remaining $k-1$ levels each get a 0/1 indicator. The choice of reference level is partly arbitrary and partly substantive: the reference becomes the baseline against which all other levels are compared, so it should be the level that makes scientific sense — placebo in a clinical trial, "no symptom" in symptom-based predictors, the largest stratum in observational studies.

Prerequisites

A working understanding of categorical variables, factor handling in R, and the role of design matrices in regression.

Theory

For a factor with k levels, treatment coding produces a $k \times (k-1)$ contrast matrix. The reference level has all-zero indicator entries; each other level has a 1 in its own column and 0 elsewhere. Substituting into the linear model:

- Intercept = mean of the reference level (in a model with only the factor as predictor).
- Coefficient on indicator j = difference $\mu_j - \mu_{\text{ref}}$.

Alternative coding schemes — sum-to-zero (`contr.sum`), Helmert (`contr.helmert`), polynomial (`contr.poly`) — change the meaning of the intercept and coefficients but produce identical fits. Ordered factors default to polynomial contrasts in R, which decompose the trend into linear, quadratic, etc. components; override with `contrasts(f) <- contr.treatment` if you want familiar group differences.

Assumptions

Same as the underlying regression: linearity in the linear predictor, etc. The choice of contrast affects coefficient interpretation but not model fit, residuals, or predictions.

R Implementation

```r
# Default: treatment contrasts with first level as reference
fit <- lm(mpg ~ factor(cyl), data = mtcars)
summary(fit)

# Change reference level
mtcars$cyl <- factor(mtcars$cyl)
mtcars$cyl <- relevel(mtcars$cyl, ref = "6")
fit2 <- lm(mpg ~ cyl, data = mtcars)
summary(fit2)
```

```r
# Inspect contrast matrix
contrasts(mtcars$cyl)
```

Output & Results

The fitted intercept equals the reference-level mean; each coefficient equals the difference from the reference. Re-leveling changes the intercept and coefficients but does not change predictions or residuals.

Interpretation

A reporting sentence: "Compared to 4-cylinder cars (reference; mean 26.7 mpg), 6-cylinder cars averaged 6.9 mpg less ($p < 0.001$) and 8-cylinder cars averaged 11.6 mpg less ($p < 0.001$). Treatment contrasts with 4-cylinder as reference were used throughout." Always state the reference level explicitly; reviewers and reanalysts cannot tell from coefficients alone.

Practical Tips

- Choose the reference level to match the scientific question — placebo, baseline, control, or "most common" — rather than relying on alphabetic default.
- Use `relevel(f, ref = "...")` or `forcats::fct_relevel()` to set a specific reference; in `lm()`/`glm()`, the first factor level is always the reference.
- For ordered factors, R defaults to polynomial contrasts (`contr.poly`); override with `contrasts(f) <- contr.treatment` if you want pairwise group differences.
- For ANOVA-style hypothesis testing, sum-to-zero contrasts (`contr.sum`) make the intercept the grand mean and produce orthogonal Type II / Type III sums of squares; specify with `options(contrasts = c("contr.sum", "contr.poly"))`.
- Inspect contrast matrices with `contrasts(f)` to confirm what is actually being fitted; mismatch between expected and actual contrasts is a common source of misinterpreted output.
- Document the contrast type in methods: "treatment contrasts with placebo as reference" leaves no room for ambiguity.

R Packages Used

Base R `factor()`, `relevel()`, `contrasts()`, `contr.treatment()`, `contr.sum()`, `contr.helmert()`, `contr.poly()`; `forcats` for tidy factor manipulation including `fct_relevel()`; `emmeans::emmeans()` for marginal means and post-hoc contrasts that are interpretation-stable across coding choices.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)

- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures $\rightarrow$ mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI